Differentiable Object Pose Connectivity Metrics for Regrasp Sequence Optimization

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***Abstract*—Regrasp planning is often required when a single pick-and-place cannot transfer an object from an initial pose to a goal pose while maintaining grasp feasibility. The main challenge is to reason about shared-grasp connectivity across intermediate poses, where discrete search becomes brittle under sparse grasp candidates. We propose an implicit multi-step regrasp planning framework based on a differentiable pose sequence cost. We model grasp feasibility under an object pose using an Energy-Based Model (EBM) and leverage energy additivity to construct a continuous energy landscape that measures pose-pair connectivity, enabling gradient-based optimization of intermediate object poses. An adaptive iterative deepening strategy is further introduced to automatically determine the minimum number of intermediate steps. Experiments show that the proposed formulation provides smooth and informative gradients, improving planning robustness over baseline connectivity objectives. We further demonstrate generalization to unseen grasp poses and cross-end-effector transfer, where a model trained with parallel-gripper constraints can guide suction-based manipulation. Finally, results under varying grasp sparsity highlight that multi-step optimization, together with sufficiently dense grasp candidates, is critical for successful regrasp planning.**

***Index Terms*—Regrasping, Manipulation planning, Deep learning in regrasping and manipulation.**

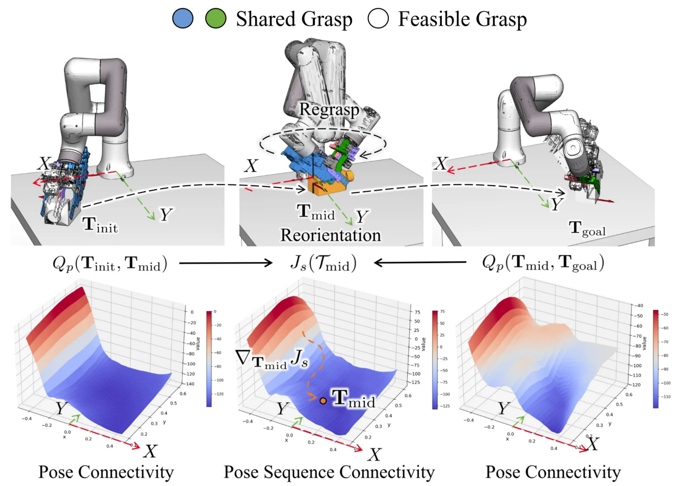

Fig. 1. When a single pick-and-place manipulation is infeasible under given object poses $\mathbf{T}_{\text{init}}$ and $\mathbf{T}_{\text{goal}}$, a regrasp via an intermediate pose $\mathbf{T}_{\text{mid}}$ (yellow) is required. We propose a *pose connectivity* score $Q_p(\mathbf{T}_a, \mathbf{T}_b)$ and a differentiable *pose sequence* cost $J_s(\mathcal{T}_{\text{mid}})$ based on EBMs to generate an optimal intermediate pose $\mathbf{T}_{\text{mid}}$ by following the gradient $\nabla_{\mathbf{T}_{\text{mid}}} J_s$. This facilitates the construction of a valid object reorientation sequence bridging the $\mathbf{T}_{\text{init}}$ and $\mathbf{T}_{\text{goal}}$, enabling multi-step pick-and-place planning.[1]

I. Introduction

In robotic pick-and-place, regrasp planning becomes necessary when no single grasp can transfer an object from an initial pose to a goal pose under robotic and environmental constraints [1]. Classic approaches address this by searching over discretized regrasp graphs constructed from sampled stable placements and grasp candidates [2][3]. However, their performance is fundamentally coupled to sampling density and discretization resolution, and planning can become brittle when the available grasp candidates are sparse or highly constrained [4][5][6]. In this paper, we overcome these drawbacks of classic approaches by proposing an implicit planning framework that replaces discrete connectivity search with differentiable connectivity metrics, enabling gradient-based intermediate-pose optimization and adaptive determination of the minimum number of regrasp steps.

At the core of regrasp planning is *shared grasp*: for each adjacent pose pair in a regrasp sequence, there must exist at least one grasp configuration that is feasible at both poses [3]. When such grasps are missing between the initial and goal poses, intermediate object poses must be introduced to "bridge" the sequence, and multiple intermediate steps may be required when feasible grasps are sparse or highly constrained. Therefore, practical regrasp planning must jointly optimize intermediate poses and find the shortest fully connected, physically feasible pose sequence.

We address these challenges with an implicit planning framework that turns shared-grasp constraints into differentiable objectives. Specifically, we model grasp feasibility with an Energy-Based Model (EBM) and derive a *pose connectivity* score by marginalizing composed grasp energies for a pose pair. This yields a smooth scalar field over the object pose space that measures shared-grasp abundance, enabling gradient-based optimization of intermediate poses. We then define a differentiable *pose sequence* cost that aggregates pairwise scores along a chain and includes a regularizer to avoid connectivity bottlenecks. Finally, we combine Langevin dynamics with an adaptive iterative deepening strategy to jointly optimize intermediate poses and search for the minimum required sequence length.

Experiments on a 6-DoF manipulator demonstrate that the proposed metric provides smooth and informative gradients.

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[1]For the 3D visualization, at each planar location (x, y) we aggregate energy over discretized in-plane orientations sampled every $30°$: $Q_p(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}) = -\alpha \sum_{\boldsymbol{\theta}^i_{\text{mid}}} \log \sum_{\boldsymbol{g} \in \mathcal{G}} \exp\Big(-(E_{\phi_f}(\mathbf{T}_{\text{init}}, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_{\text{mid}}, \boldsymbol{g}))/\alpha \Big)$, which serves as the scalar value plotted at that (x, y) location.

As a result, planning becomes more reliable compared with baseline connectivity formulations and generalizes to unseen grasp poses. We further show cross-end-effector transfer, where a model trained with parallel-gripper constraints can guide suction-based manipulation.

In summary, our key contribution is a differentiable pose and sequence connectivity metric derived from compositional EBMs, which transforms discrete shared-grasp constraints into a smooth energy landscape and enables gradient-based optimization of intermediate object poses. Building on this metric, we also present an instantiation as an adaptive multi-step planner that jointly optimizes intermediate poses and determines the required number of regrasp steps via iterative deepening and Langevin dynamics.

II. Related Work

A. Graph-based Regrasp Planning

Graph-based regrasp planning typically constructs a connectivity graph where nodes represent discretely sampled object-grasp pairs. Edges are established to connect nodes sharing the same object pose (Transit actions) or the same grasp configuration across different object poses (Transfer actions). Given initial and target object poses, planning algorithms are then employed to evaluate connectivity and search for feasible paths across grasp modalities. Wan et al. [7] pioneered the application of this approach to single-arm regrasp and dual-arm handover planning in tabletop settings. Addressing environmental complexity, Ma et al. [8] extended this framework to unstructured and cluttered static environments. Beyond standard pick-and-place tasks, graph search methods have been widely applied to dexterous manipulation scenarios. Cruciani et al. [9] introduced graph search to in-hand manipulation, addressing regrasp planning where fingers cannot maintain continuous contact. Building on this, Nagahama et al. [10] utilized graph search to mitigate uncertainty in in-hand manipulation, significantly enhancing grasp precision. Furthermore, to tackle planning challenges involving complex dynamics or long horizons, researchers have introduced additional modalities and abstraction strategies. Considering grasp stability, Hu et al. [11] incorporated sliding actions into the regrasp framework to expose potentially feasible grasps, establishing a multi-modal transition graph planned via Markov Decision Processes (MDPs). In deformable object assembly tasks, Qin et al. [12] proposed constructing a local Voronoi graph around the object's ideal path, treating regrasping as a passive repair mechanism to resolve geometric blockages. For sequential manipulation puzzles involving rigid bodies, Levit et al. [13] proposed reducing planning dimensionality by searching for sequences of easier sub-problems, using heuristics to guide the operation order. Furthermore, Levit and Toussaint [14] introduced Regrasp Maps, which voxelize space and construct a global state abstraction based on grasp signatures, effectively guiding solvers to discover complex regrasp sequences in highly constrained environments.

However, explicit discrete graphs suffer from a trade-off between resolution and efficiency, risking the loss of solution space. Meanwhile, continuous sampling methods (e.g., Rapidly Exploring Random Tree (RRT)) rely on uninformed exploration, where the strict simultaneous connectivity required for 'Init-Mid-Goal' chains creates narrow passages at the intersection of disjoint grasp regions, and finding these rare intersections via blind sampling is computationally inefficient. Most critically, neither paradigm offers a differentiable landscape; they rely on heuristic search over discrete states rather than following a clear gradient signal to the solution.

To bridge this gap, we propose an implicit planning framework that composes a continuous energy landscape. By modeling grasp feasibility as an EBM, individual constraints are composed into a combinatorial gradient field. This enables explicit computation of the derivative with respect to the object pose, providing a dense, informative signal derived from the learned grasp manifold that actively guides optimization into narrow feasible regions. Consequently, this formulation provides a smoother optimization landscape compared to often discontinuous geometrical constraints, while offering gradient-based guidance typically unavailable in discrete graph search or blind sampling approaches.

B. Learning-based Regrasp Planning

Data-driven methods have made significant progress in generating such poses. Wada et al. [15] learned to generate feasible reorientation waypoints for visual manipulation, while Mishra and Chen [16] utilized diffusion models to sample intermediate poses between start and goal configurations. For multi-step scenarios, Xu et al. [17] proposed a hierarchical planning framework that guides reorientation search by learning a path cost estimator. However, this approach largely remains a heuristic extension of single-step search and lacks global optimization from a sequential perspective. Recent works have also focused on predicting diverse stable placements directly from point clouds [18] or constructing incremental regrasp graphs via closed-loop perception [19]. Nevertheless, a fundamental trade-off exists between generalization to unknown objects and the precision of physical constraints. While approaches aiming for category-level generalization excel at handling novel geometries, they often necessitate a relaxation of physical constraints to accommodate shape variance. In multi-step manipulation, however, even minor deviations from the stable manifold can accumulate, leading to failure in long-horizon tasks. Existing methods primarily focus on single-step or local reactive planning, leaving the global optimization for multi-step sequences under sparse connectivity largely unaddressed.

In contrast, in this work, we address this challenge by making a deliberate design choice: we prioritize execution robustness over generalization to unknown categories. By explicitly leveraging the known geometry assumption, we trade generalization to unknown instances for the strict enforcement of stability constraints. We analytically constrain poses to the exact $SE(2)$ stable manifolds, eliminating physical instability by definition. This allows us to reframe the problem not as a loose approximation, but as a precise iterative optimization on a continuous Free Energy landscape, enabling the autonomous determination of the optimal sequence length.

III. Energy-Based Regrasp Planning

This section presents our energy-based formulation for multi-step regrasp planning. We first model grasp feasibility and shared grasps using EBM, then derive differentiable pose and sequence connectivity metrics that enable gradient-based optimization of intermediate object poses. Finally, we describe the adaptive iterative deepening planner used to determine the required number of regrasp steps.

A. Shared Grasp

Let Σ_{W}, Σ_{O}, and Σ_{E} denote the world frame, the object's canonical frame, and the end-effector frame, respectively. The 6D object pose in the world frame is represented by ${}^{\Sigma_{\mathrm{W}}}\mathbf{T}_{\Sigma_{\mathrm{O}}} \in SE(3) \subset \mathbb{R}^{4\times4}$. A grasp configuration is parameterized as a tuple $\boldsymbol{g} = ({}^{\Sigma_{\mathrm{O}}}\mathbf{T}_{\Sigma_{\mathrm{E}}}, w)$ (Fig. 2(a-c)), where ${}^{\Sigma_{\mathrm{O}}}\mathbf{T}_{\Sigma_{\mathrm{E}}} \in SE(3)$ represents the end-effector pose in the object canonical frame and $w \in [0,1]$ denotes the gripper opening normalized by its maximum stroke. Accordingly, we define the grasp space in the object frame as $\mathcal{G} \triangleq SE(3) \times [0,1]$, with $\boldsymbol{g} \in \mathcal{G}$. Since grasps are represented in Σ_{O}, $\mathcal{G}$ is pose-invariant. For any object pose ${}^{\Sigma_{\mathrm{W}}}\mathbf{T}_{\Sigma_{\mathrm{O}}}$ and grasp $\boldsymbol{g} \in \mathcal{G}$, the absolute end-effector pose in the workspace is given by ${}^{\Sigma_{\mathrm{W}}}\mathbf{T}_{\Sigma_{\mathrm{E}}} = {}^{\Sigma_{\mathrm{W}}}\mathbf{T}_{\Sigma_{\mathrm{O}}} \cdot {}^{\Sigma_{\mathrm{O}}}\mathbf{T}_{\Sigma_{\mathrm{E}}}$ (Fig. 2(d)). For notational simplicity, we omit the frame superscripts/subscripts henceforth, as the reference frames are clear from context.

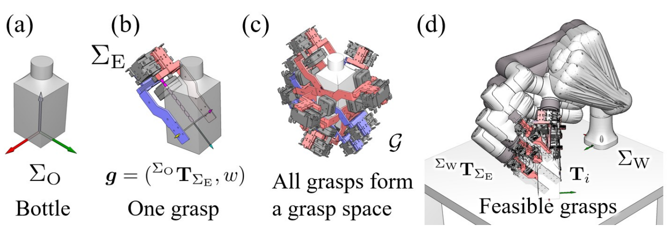

Fig. 2. (a) Bottle object. (b) An annotated grasp defined in the object canonical frame. (c) All grasps form a grasp space $\mathcal{G}$. (d) Feasible grasps in the world frame, or namely workspace.

To evaluate grasp feasibility, we learn an energy function $E_{\phi_f} : SE(3) \times \mathcal{G} \rightarrow \mathbb{R}$, where ϕ_f denotes the learned parameters of the feasibility energy model. The inputs to the model include the object pose $\mathbf{T}$ and a grasp $\boldsymbol{g} \in \mathcal{G}$, and the output is a scalar energy value. Lower energy values indicate a higher likelihood that $\boldsymbol{g}$ is feasible under $\mathbf{T}$ with respect to IK solvability and collision constraints.

A grasp is defined as a *shared grasp* between an initial object pose $\mathbf{T}_{\mathrm{init}}$ and a target pose $\mathbf{T}_{\mathrm{goal}}$ if the grasp configuration is feasible at both poses [20][21]. We model the prediction of shared grasps as a joint probability estimation problem. Given $\mathbf{T}_{\mathrm{init}}$ and $\mathbf{T}_{\mathrm{goal}}$, our objective is to identify grasps $\boldsymbol{g} \in \mathcal{G}$ that lead to a high joint probability $p(\mathbf{T}_{\mathrm{init}}, \mathbf{T}_{\mathrm{goal}}, \boldsymbol{g})$. This joint distribution can be approximated as the product of two independent terms:

$$p(\mathbf{T}_{\mathrm{init}}, \mathbf{T}_{\mathrm{goal}}, \boldsymbol{g}) \propto p(\mathbf{T}_{\mathrm{init}}, \boldsymbol{g}) \cdot p(\mathbf{T}_{\mathrm{goal}}, \boldsymbol{g}). \quad (1)$$

Following the Boltzmann distribution $p(\boldsymbol{x}) \propto \exp(-E(\boldsymbol{x}))$, the approximation yields an additive energy form:

$$p(\mathbf{T}_{\mathrm{init}}, \mathbf{T}_{\mathrm{goal}}, \boldsymbol{g}) \propto \exp\left(-\left[E_{\phi_f}(\mathbf{T}_{\mathrm{init}}, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_{\mathrm{goal}}, \boldsymbol{g})\right]\right). \quad (2)$$

This implies that we can efficiently evaluate the joint probability, and thus shared grasps, by summing the energy of a grasp $\boldsymbol{g}$ under $\mathbf{T}_{\mathrm{init}}$ and $\mathbf{T}_{\mathrm{goal}}$ [22], [23].

All shared grasps under given initial and goal poses form a subset $\mathcal{G}_{\mathrm{shared}} \subset \mathcal{G}$ where the summed energy falls below a certain threshold h [24]:

$$\mathcal{G}_{\mathrm{shared}} = \{\boldsymbol{g} \in \mathcal{G} \mid E_{\phi_f}(\mathbf{T}_{\mathrm{init}}, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_{\mathrm{goal}}, \boldsymbol{g}) \leq h\}. \quad (3)$$

Fig. 3 illustrates this energy additivity: individual energy landscapes assign low energy to feasible grasps under each pose, and summing them yields a joint energy whose minima align with grasps feasible in both configurations.

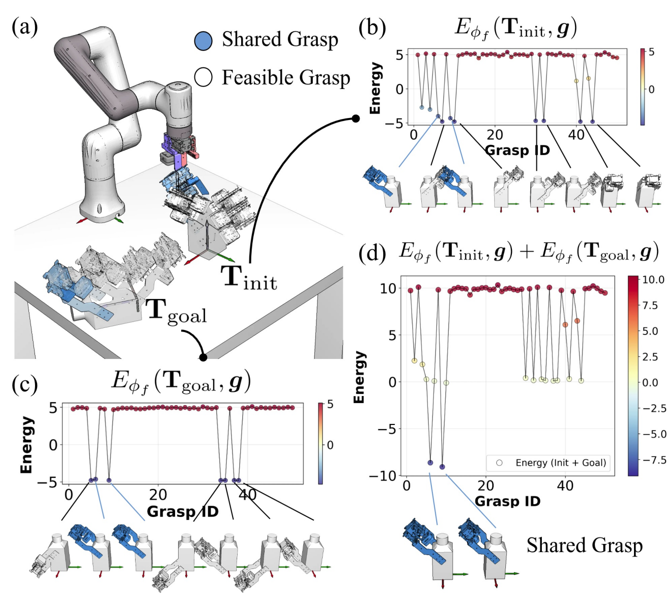

Fig. 3. Validation of energy compositionality. (a) Ground truth feasible (white) and shared (blue) grasps. (b)-(c) Individual energy landscapes E_{ϕ_f} for initial and goal poses. (d) Joint energy distribution obtained by summation, where low-energy minima correctly align with the shared grasps in (a).

B. Object Pose Connectivity Score

Equation (3) provides a set-valued characterization of shared grasps. When $\mathcal{G}_{\mathrm{shared}} = \emptyset$, a direct transfer from $\mathbf{T}_{\mathrm{init}}$ to $\mathbf{T}_{\mathrm{goal}}$ under a single grasp becomes infeasible, and regrasping via intermediate object poses is required. To enable gradient-based search in the object pose space, we define a scalar score to quantify the shared-grasp connectivity between two poses. Specifically, we use a *pose connectivity* score function $Q_p(\mathbf{T}_a, \mathbf{T}_b) : SE(3) \times SE(3) \rightarrow \mathbb{R}$ to reflect the existence and abundance of shared grasps. Following equation (2), $Q_p(\mathbf{T}_a, \mathbf{T}_b)$ can be approximated by marginalizing the additive grasp energy over the grasp space: $\mathcal{G}_{\mathrm{shared}}$干什么用？注意shared不是斜体 在Fig.5(c)有$|\mathcal{G}_{\mathrm{shared}}|$表示mid和init、goal各自的共享抓取数目，$|\mathcal{G}_{\mathrm{shared}}|_s$表示mid和init、goal同时有共享抓取时，Mid-init, mid-goal共享抓取数目之和，一种从sequence的

角度去衡量的。已在Fig. 5的Caption部分解释。

$$
\begin{aligned}
&Q_p(\mathbf{T}_a, \mathbf{T}_b) \\
&= -\alpha \log \int_{\boldsymbol{g}\in\mathcal{G}} \exp\left(-\frac{E_{\phi_f}(\mathbf{T}_a, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_b, \boldsymbol{g})}{\alpha}\right) d\boldsymbol{g} \\
&\approx -\alpha \log \sum_{\boldsymbol{g}\in\mathcal{G}} \exp\left(-\frac{E_{\phi_f}(\mathbf{T}_a, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_b, \boldsymbol{g})}{\alpha}\right), \qquad (4)
\end{aligned}
$$

where α is the scaling factor controlling distribution smoothness. In practice, we pre-sample a discrete set of M candidate grasps from $\mathcal{G}$, denoted by $\mathcal{G} = \{\boldsymbol{g}_j\}_{j=1}^{M} \subset \mathcal{G}$ [25], to approximate the integral by the finite summation.

Using Q_p, we can evaluate end-to-end regrasp feasibility along an object pose chain. Given a pose sequence $\mathbf{T}_{\text{init}}, \{\mathbf{T}\}_{i=1}^{N}, \mathbf{T}_N, \mathbf{T}_{\text{goal}}$, we define a differentiable *pose sequence* cost

$$
\begin{aligned}
J_s(\{\mathbf{T}\}_{i=0}^{N+1}) = &\underbrace{\sum_{i=0}^{N} Q_p(\mathbf{T}_i, \mathbf{T}_{i+1})}_{\text{Connectivity Term}} + \\
&\underbrace{\lambda_{\text{reg}} \sum_{i=0}^{N-1} \left(Q_p(\mathbf{T}_i, \mathbf{T}_{i+1}) - Q_p(\mathbf{T}_{i+1}, \mathbf{T}_{i+2})\right)^2}_{\text{Regularization Term}}, \qquad (5)
\end{aligned}
$$

with $\mathbf{T}_0 \coloneqq \mathbf{T}_{\text{init}}$ and $\mathbf{T}_{N+1} \coloneqq \mathbf{T}_{\text{goal}}$ for simplification. $\{\mathbf{T}\}_{i=1}^{N}$ denotes the intermediate pose sequence to be optimized. The cost is differentiable with respect to the intermediate poses, enabling gradient-based optimization. The first term of the definition encourages high shared-grasp connectivity across all adjacent pose pairs in the sequence. The second term is a regularizer that promotes uniform connectivity distribution along the sequence, preventing bottlenecks where one transition is significantly more difficult than others.

When $N = 0$, the chain reduces to $(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}})$ and the objective becomes $J_s(\emptyset) = Q_p(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}})$, corresponding to a direct pick-and-place without intermediate poses. When $N = 1$, the chain reduces to $(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}})$ and $J_s(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}})$ corresponds to pick-and-place with one time of intermediate regrasp.

C. Multi-step Regrasp Planning

Our goal is to find the minimum number of intermediate poses N and the corresponding pose sequence such that every adjacent pose pair along the chain admits at least one shared grasp. We achieve this by converting the set-valued shared-grasp constraint into a differentiable objective: the pairwise pose connectivity score Q_p evaluates shared-grasp abundance between two poses, and the pose sequence cost J_s aggregates these terms over the full chain (equation (5)). Since N is unknown, we adopt an adaptive iterative deepening planner (Algorithm 1) that increases N from small to large; for each N, it optimizes a batch of candidate sequences by minimizing J_s, then accepts the first N whose best K_{top} candidate passes shared-grasp verification on every adjacent pose pair. In practice, the Step 3 verification can be accelerated by performing batch verification over the top-K_{top} candidates, as shown in Algorithm 2.

Algorithm 1: Adaptive Iterative Deepening Planning

for $N = 1$ **to** $N_{\max}$ **do**
  // 1. Initialization
  Initialize batch $\boldsymbol{\Psi}_B = \{\{\mathbf{T}\}_{i=1}^{N\,(b)}\}_{b=1}^{B}$ via equation (6) ;
  // 2. Langevin Dynamics
  for $k = 1$ **to** K_{opt} **do**
    Compute $\nabla_{\{\boldsymbol{\xi}_i\}_{i=1}^{N}} J_s$ for all $\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_B$;
    Update $\{\boldsymbol{\xi}_i\}_{i=1}^{N}$ using equation (7) ;
  // 3. Sequential Verification
  $\{\mathbf{T}\}_{i=1}^{N\,(*)} \leftarrow \operatorname{argmin}_{\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_B} \left(J_s(\{\mathbf{T}\}_{i=1}^{N})\right)$;
  $\{\mathbf{T}\}_{i=0}^{N+1} \leftarrow \{\mathbf{T}_{\text{init}}, \{\mathbf{T}\}_{i=1}^{N}, \mathbf{T}_{\text{goal}}\}$;
  is_valid $\leftarrow$ **True** ;
  for $i = 0$ **to** N **do**
    Compute shared set: $\mathcal{S}_i \leftarrow$
      $\{\boldsymbol{g} \in \mathcal{G}_{\text{ref}} \mid E_{\phi_f}(\mathbf{T}_i, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_{i+1}, \boldsymbol{g}) \leq h\}$;
    if $\mathcal{S}_i = \emptyset$ **then**
      is_valid $\leftarrow$ **False** ;
      break ;
  if *is_valid* **then**
    return $\{\mathbf{T}\}_{i=1}^{N\,(*)}$
return Failure ;

Algorithm 2: Batch Verification for Step 3 (Practical Variant)

$\boldsymbol{\Psi}_{\text{top}} \leftarrow \operatorname{argmin}^{K_{\text{top}}}_{\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_B} \left(J_s(\{\mathbf{T}\}_{i=1}^{N})\right)$;
for $i = 0$ **to** N **do**
  Compute shared set:
    $\mathcal{S}_i \leftarrow \{\boldsymbol{g} \in \mathcal{G}_{\text{ref}} \mid E_{\phi_f}(\mathbf{T}_i, \boldsymbol{g}) + E_{\phi_f}(\mathbf{T}_{i+1}, \boldsymbol{g}) \leq h\}$
    for all $\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_{\text{top}}$;
  Update candidates: $\boldsymbol{\Psi}_{\text{top}} \leftarrow \{\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_{\text{top}} \mid \mathcal{S}_i \neq \emptyset\}$;
  if $\boldsymbol{\Psi}_{\text{top}} = \emptyset$ **then**
    return $\emptyset$;
return $\operatorname{argmin}_{\{\mathbf{T}\}_{i=1}^{N} \in \boldsymbol{\Psi}_{\text{top}}} \left(J_s(\{\mathbf{T}\}_{i=1}^{N})\right)$;

Here, we assume the object has a known shape and is placed in a tabletop environment. Intermediate poses are constrained to planar perturbations around canonical stable placements $\{\mathbf{T}_{\text{s}}^{(a)}\}_{a=1}^{S}$. Each intermediate pose $\mathbf{T}_i$ (with $i = 1, \ldots, N$) is parameterized by selecting a stable placement index $m \in \{1, \ldots, S\}$ and applying planar parameters $\boldsymbol{\xi}_i = [x_i, y_i, \theta_i]^\top$:

$$
\mathbf{T}_i = \Delta\mathbf{T}(\boldsymbol{\xi}_i)\,\mathbf{T}_{\text{s}}^{(m)}, \quad \Delta\mathbf{T}(\boldsymbol{\xi}_i) \coloneqq \begin{bmatrix} \mathbf{R}_z(\theta_i) & [x_i, y_i, 0]^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}, \qquad (6)
$$

where $\mathbf{R}_z(\theta_i)$ denotes a yaw rotation about the world z-axis. Fig. 4 illustrates the constraints. Intermediate sequences are initialized by applying planar perturbations to randomly sampled stable placement sequences.

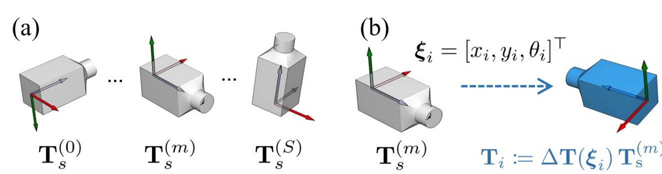

Fig. 4. (a) Stable placements on a tabletop environment. (b) Intermediate poses constrained to planar perturbations and rotations.

To perform optimization, we generate a batch $\boldsymbol{\Psi}_B$ containing B sequences of length N using the table top sampling and perturbation. For each particular N, we minimize $J_s(\{\mathbf{T}\}_{i=0}^{N+1})$ using Langevin dynamics with the following update rule:

$$\boldsymbol{\xi}_i^{(k+1)} \leftarrow \boldsymbol{\xi}_i^{(k)} - \eta \nabla_{\boldsymbol{\xi}_i} J_s(\{\mathbf{T}\}_{i=0}^{N+1^{(k)}}) + \sqrt{2\eta\tau} \cdot \boldsymbol{z}_k, \quad (7)$$

where $\boldsymbol{z}_k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ is the random noise term, η is the step size, k is the iteration step, and τ is the temperature parameter. The noise term regulates the balance between local gradient guidance and global exploration, producing optimized candidate sequence $\{\mathbf{T}\}_{i=0}^{N+1^{(*)}}$ for the fixed N.

IV. Experiments and Analysis

We conducted experiments on a 6-DoF Dobot Nova2 manipulator with a parallel gripper, using an Intel Core i9-13900KF (128GB RAM) with an RTX 4090 GPU. The workspace was constrained to $x \in [-0.45, 0.45]$ m, $y \in [0.1, 0.6]$ m, and $\theta \in [0, 2\pi]$ rad. The EBM, a three-layer MLP with SELU activations, was trained on 20,000 feasible grasp samples (details in Appendix V-B). Optimization used $K_{\text{opt}} = 20$ iterations with step size $\eta = 0.3$ and noise $\tau = 0.1$. Connectivity parameters were $\alpha = 1.0$ and $\lambda_{\text{reg}} = 0.5$, with the energy threshold h calibrated individually per object. For fair comparison, we set the batch size to $B = 200$ for each stable placement during single-step generation, and scaled it to $B = 1000$ random sequences for multi-step planning.

The experimental evaluation consisted of four parts. First, we conducted an ablation study on different connectivity formulations to analyze gradient properties. Second, we compared our method against a discrete search baseline obtained by exhaustive grid sampling for single-step intermediate pose generation. Third, we evaluated generalization to unseen grasp poses and transferability across different end-effectors. Finally, we evaluated adaptive multi-step regrasp planning performance under varying grasp sparsity levels.

A. Ablation Study of Connectivity Formualtions

We first analyze how different connectivity formulations shape the optimization landscape. The goal is to verify that the proposed pose sequence cost J_s yields informative gradients while avoiding two common failure modes: spurious local minima caused by asymmetric connectivity in naive summation objectives, and vanishing gradients induced by hard truncation.

To isolate the effect of the objective from optimizer stochasticity, we fix a challenging pose pair $(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}})$ and sweep the intermediate pose $\mathbf{T}_{\text{mid}}$ along a controlled trajectory: a 40-step linear interpolation from $\mathbf{T}_{\text{goal}}$ to $\mathbf{T}_{\text{init}}$ (Fig. 5(a)). Along this trajectory, we evaluate J_s and compare it with two baselines: a truncated objective (Appendix V):

$$J_s^h = Q_p^h(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}) + Q_p^h(\mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}}) + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}, \quad (8)$$

and a naive summation objective without the regulator:

$$J_s^+ = Q_p(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}) + Q_p(\mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}}). \quad (9)$$

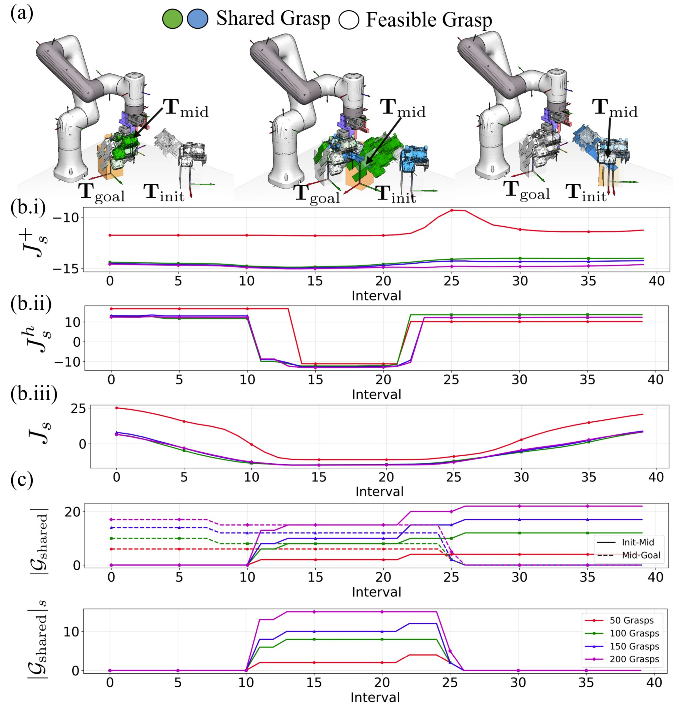

Fig. 5. Evaluation of pose sequence costs via pose interpolation in various grasp candidates. (a) Interpolated object motion from Goal to Start ($\mathbf{T}_{\text{mid}}$). (b) Comparison of energy landscapes: (i) Naive summation J_s^+ exhibits spurious minima due to asymmetric connectivity; (ii) Truncated objective J_s^h suffers from flat surfaces (vanishing gradients); (iii) Proposed cost J_s ($\lambda_{\text{reg}} = 0.5$) shows a distinct global minimum with strong gradients. (c) Ground truth count of shared grasps, confirming that J_s correctly identifies the valid regrasp region.where $|\mathcal{G}_{\text{shared}}|$ denotes the number of shared grasps for individual pose pairs ($(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}})$ or $(\mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}})$). Furthermore, $|\mathcal{G}_{\text{shared}}|_s$ represents the sum of shared grasps across the entire sequence $(\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}})$, counted only when both transitions are simultaneously feasible.

Results are shown in Fig. 5, the Fig. 5(b.i) reveals the flaw in the naive summation J_s^+: it masks asymmetric connectivity distributions, leading to spurious minima where physical connectivity is broken. Fig. 5(b.ii) shows that J_s^h creates flat potential surfaces in non-connected regions, causing vanishing gradients that hinder optimization. Conversely, Fig. 5(b.iii) demonstrates the necessity of the energy-variance regularizer in J_s: it eliminates spurious minima and reshapes the energy landscape into a convex structure with significant gradients, ensuring the optimizer effectively converges to a solution where shared grasps exist across the entire $\{\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{mid}}, \mathbf{T}_{\text{goal}}\}$ sequence.

B. Comparison with Discrete Search Baseline

To validate the physical feasibility of the generated intermediate poses, we designed a single-step reorientation benchmark and compared the results against Ground Truth (GT). Based on the stable placement manifolds defined in Section III-C, we established feasible GT regions by performing exhaustive grid sampling on the $SE(2)$ subspace (with a spatial resolution of 0.01 m and an angular resolution of 60°) and filtering candidates via the pose sequence cost check. The experiment utilized four representative objects (Bottle, Bunny, Pentagon, Mug), each with 200 grasp candidates, tested across 10 sampled pose pairs $\{\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}}\}$ that lack shared grasps but are guaranteed to be solvable via a single intermediate transition which at least have a mid pose have shared grasp with $\mathbf{T}_{\text{init}}$ and $\mathbf{T}_{\text{goal}}$, meanwhile we set the sequence length $N_{\max} = 1$ in the optimization process. We benchmarked our proposed cost J_s against the naive cost J_s^+ and the truncated cost J_s^h. For evaluation, we perform the batch verification as shown in Algorithm 2, and report the average validity rates defined as the percentage of generated $\mathbf{T}_{\text{mid}}$ that share grasps with both $\mathbf{T}_{\text{init}}$ and $\mathbf{T}_{\text{goal}}$ for $K_{\text{top}} = 200$, 100, and 10.

Fig. 6(b) illustrates a representative optimization instance for the bottle object, which has five stable planar placements as shown in Fig. 6(a). Fig. 6(c) shows that the intermediate poses $\mathbf{T}_{\text{mid}}$ generated by the proposed method (c.iv)删除 exhibit a high degree of alignment with the GT feasible regions shows in (c.i). 没有任何信息解释箭头，颜色都是什么意思好的，我在(c.i)中解释 In comparison, the naive metric (c.ii) and the truncated metric (c.iii) appear less effective in clustering particles within valid regions. Table I shows the average validity rates. The results show that the proposed method generally provides improved validity rates across the tested scenarios, particularly in identifying the broad feasible solution space. This trend is particularly notable for the Mug, where the complex geometry results in narrow connectivity passages that may pose challenges for baseline metrics. While baseline metrics can show competitive precision within the narrow K_{top}=10 range for certain shapes (e.g., Naive on Bunny), their performance tends to decrease in broader sampling regimes. The observed robustness likely stems from the continuous gradient guidance, which helps mitigate the risk of particles becoming trapped in zero-gradient plateaus (typical of J_s^h) or spurious asymmetric minima (typical of J_s^+).

TABLE I
COMPARISON OF AVERAGE VALIDITY RATE

Object	$K_{\text{top}} = 200$	$K_{\text{top}} = 100$	$K_{\text{top}} = 10$
Bottle	83.1 / 73.0 / 51.7	85.9 / 75.3 / 58.0	82.0 / 77.0 / 60.0
Mug	70.8 / 24.9 / 29.9	83.7 / 37.6 / 53.0	76.0 / 43.0 / 42.0
Bunny	56.0 / 53.0 / 37.2	55.3 / 54.6 / 38.3	48.0 / 57.0 / 32.0
Pentagon	67.0 / 56.5 / 65.4	63.6 / 56.0 / 62.5	60.0 / 51.0 / 61.0

Note 1 J_s (Proposed) / J_s^+ (Naive) / J_s^h (Truncated).
Note 2 All results report the average validity across 10 $\{\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}}\}$ pose pairs.validity是用在pair上的？不是整个pose上吗？可能这里没有表述清楚，这里是所有的结果是在采样到的10个$\{\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}}\}$对进行$\mathbf{T}_{\text{mid}}$规划的平均结果。validity rate是什么意思？谁的valid 的rate? Ktop是取的Js最小的topK? 和valid有什么关系 validity rate是生成的K_{top}个$\mathbf{T}_{\text{mid}}$中同时跟$\mathbf{T}_{\text{init}}$和$\mathbf{T}_{\text{goal}}$具有共享抓取的占比；似乎没有清晰的解释，我在上文补充解释，后文的实验中也是这样定义的。

C. Generalization and Transferability

1) Mid-pose generation on unseen grasps: In this experiment, we evaluated the generalization capability of the J_s cost using a separate randomly sampled set of 200 unseen grasps in 10 given $\{\mathbf{T}_{\text{init}}, \mathbf{T}_{\text{goal}}\}$ pairs same as experiment IV-B.

TABLE II
GENERALIZATION PERFORMANCE ON UNSEEN GRASPS

Object	K_{top}=200	K_{top}=100	K_{top}=50	K_{top}=10
Bottle	83.1 / 72.9	85.9 / 72.8	85.4 / 72.2	82.0 / 76.0
Mug	70.8 / 16.1	83.7 / 28.6	82.4 / 34.8	76.0 / 29.0
Bunny	56.0 / 45.8	55.3 / 49.8	53.2 / 51.6	48.0 / 59.0
Pentagon	67.0 / 41.4	63.6 / 46.0	63.2 / 45.6	60.0 / 45.0

Note Values are the average validity rates across 10 pose pairs: Seen (Proposed) / Unseen. 为什么表1和表2的Ktop数量不同 感觉放不下了，所以就删掉一个...

As presented in Table II, the method exhibits a capability to generalize to unseen grasp configurations, though with an expected performance gap compared to the training set (Seen). For regular geometries like the Bottle, the validity rate on unseen grasps remains robust. Interestingly, for the Bunny, the method achieves higher precision in the narrow K_{top}=10 range for unseen grasps. This suggests that the learned energy landscape captures the underlying object geometry effectively, allowing the optimizer to discover highly feasible regions within the unseen set that were not explicitly covered during training. However, for objects like the Mug and the Pentagon, a more significant drop is observed. This gap may stem from: (1) insufficient training samples (200 grasps) to capture geometric diversity, limiting EBM generalization; (2) sampled unseen grasps inadequately covering regions necessary for successful transitions. Both factors can lead to biased energy evaluations and reduced planning success.

2) Cross-End-Effector Evaluation: To investigate the versatility of the EBM framework, we evaluated its generalization capability across diverse hardware configurations. Six end-effectors (EEs) were classified based on the strictness of their geometric constraints: three parallel grippers (WRS3, Robotiq-140, and Robotiq-85) representing stringent constraints (requiring dual-contact alignment and stroke clearance), and three suction cups representing relaxed constraints (requiring only single-surface contact), with varying Tool Center Point (TCP) offsets across both categories. We conducted a 6×6 cross-evaluation where an EBM trained on EE A was employed to guide mid-pose optimization for a target EE B. To ensure a fair comparison, all EBMs were trained on datasets of equal size (20k samples). We reported the validity rate within the top 100 generated poses, evaluated on randomly sampled object pose pairs, to quantify how effectively the learned energy landscape transfers between hardware with differing physical priors.

Our 6×6 cross-evaluation reveals a predominantly unidirectional transferability governed by constraint strictness. EBMs trained on strict-constraint hardware (parallel grippers) effectively guide planning for relaxed-constraint hardware (suction cups) by acting as a conservative manifold prior, with validity rates typically exceeding 70%. Conversely, models trained on suction cups generally fail to generalize to parallel grippers (success rates often below 40%), with limited exceptions. This suggests a Manifold Subset relationship: the EBM captures a fundamental graspability prior where the feasible space of strict end-effectors is a subset of the relaxed ones. Thus, training on the most restrictive hardware yields a versatile

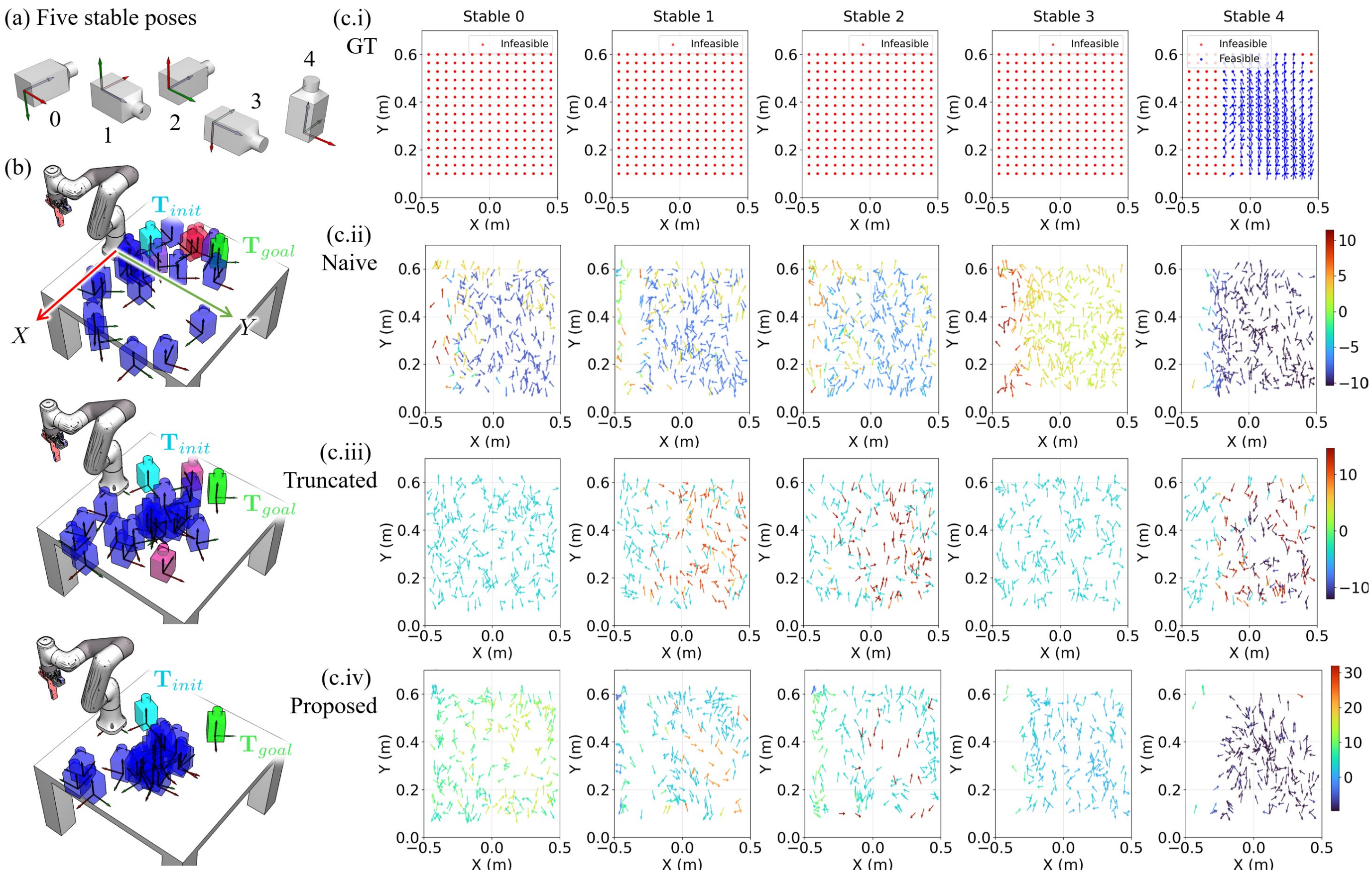

Fig. 6. Optimization ~~process~~result with the bottle object. (a) Five stable poses. (b) Evolution of the energy landscape during optimization (Blue: low energy/high connectivity). (c) Comparison of generated object pose distributions, the arrow provides a 2D visualization of each intermediate pose $\mathbf{T}_{\mathrm{mid}}$, originating at (x, y) and pointing along the planar yaw θ in $\boldsymbol{\xi} = [x, y, \theta]^{\top}$, i.e., $(\cos\theta, \sin\theta)$; arrow color indicates pose cost (energy), with red for high energy and blue for low energy, consistent with the color bar. In the actual evaluation, the full 3D pose is used, with z-height and base orientation from the stable placement $\mathbf{T}_{\mathrm{s}}^{(m)}$ and in-plane rotation $\mathbf{R}_z(\theta)$ (Eq. 6).: (i) Ground Truth (GT) feasible regions obtained via exhaustive grid sampling; (ii-iii) Results from the naive cost J_s^+ and the truncated cost J_s^h, respectively. (iv) Results of the proposed cost J_s.每个图是优化的第几步？为什么是pose相关，箭头方向表示什么？颜色表示能量应明确说明。 1. (c)的图都是优化的结果，即最后一步。2.好的，关于箭头和能量的颜色的说明在图中的caption增加了。

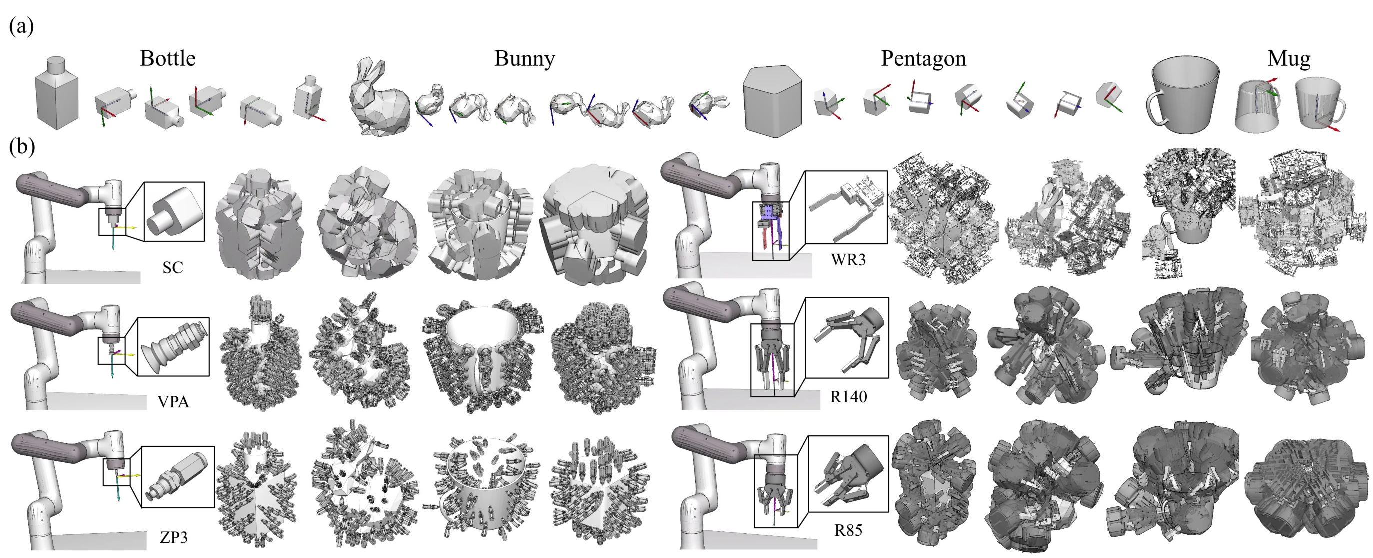

Fig. 7. Dataset overview: (a) Four representative object shapes (Bottle, Bunny, Pentagon, Mug) and their table-stable placement poses. (b) Pre-sampled grasp candidates set $\mathcal{G}$ for six different end-effectors (three parallel grippers: WRS3, Robotiq-140/85; three suction cups: SC, VPA, ZP3) on different objects, used for cross-validation experiments to evaluate transferability across hardware configurations.这个图是要如何修改来着？机器人的存在是否有意义？手抓不同比例尺是误导。 把手爪的图片进行了修改，用机械臂来做参照物来看各个末端执行器放的相对大小情况。

TABLE III
CROSS-VALIDATION VALIDITY RATES

		W	R1	R2	S1	S2	S3			W	R1	R2	S1	S2	S3
Bt	W	100	100	100	82.0	77.0	83.0	Bn	W	100	0.0	93.0	40.0	43.0	46.0
	R1	89.0	98.0	98.0	93.0	93.0	95.0		R1	31.0	100	95.0	100	94.0	98.0
	R2	82.0	100	99.0	69.0	93.0	94.0		R2	89.0	0.0	84.0	30.0	34.0	32.0
	S1	40.0	92.0	76.0	89.0	74.0	79.0		S1	11.0	0.0	17.0	98.0	92.0	76.0
	S2	61.0	94.0	92.0	67.0	88.0	72.0		S2	25.0	0.0	17.0	94.0	90.0	30.0
	S3	4.0	70.0	69.0	86.0	82.0	88.0		S3	0.0	0.0	0.0	39.0	98.0	82.0
M	W	94.0	72.0	90.0	100	90.0	95.0	P	W	90.0	92.0	88.0	77.0	52.0	64.0
	R1	14.0	0.0	52.0	32.0	73.0	79.0		R1	78.0	100	83.0	73.0	82.0	95.0
	R2	54.0	88.0	99.0	70.0	76.0	84.0		R2	78.0	84.0	80.0	92.0	93.0	99.0
	S1	0.0	0.0	0.0	100	100	96.0		S1	8.0	24.0	0.0	85.0	86.0	75.0
	S2	0.0	0.0	5.0	100	100	100		S2	39.0	59.0	30.0	85.0	83.0	65.0
	S3	0.0	0.0	0.0	100	100	100		S3	5.0	31.0	18.0	90.0	81.0	89.0

Note 1 Tr/Te: Training device / Testing device.
Note 2 W: WRS3; R1: Robotiq-85; R2: Robotiq-140.
Note 3 Lime cells indicate the same device for training and testing.

planner capable of generalizing to less-constrained devices.

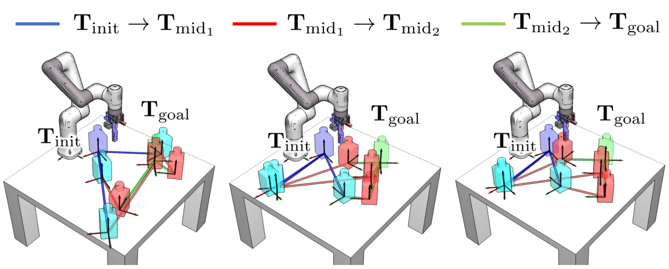

Fig. 8. Evolution of multi-step trajectory optimization (left to right). Cyan and red meshes represent $\mathbf{T}_{\text{mid}_1}$ and $\mathbf{T}_{\text{mid}_2}$. Colored lines denote sequential transitions: Blue ($\mathbf{T}_{\text{init}} \rightarrow \mathbf{T}_{\text{mid}_1}$), Red ($\mathbf{T}_{\text{mid}_1} \rightarrow \mathbf{T}_{\text{mid}_2}$), and Green ($\mathbf{T}_{\text{mid}_2} \rightarrow \mathbf{T}_{\text{goal}}$).

D. Evaluation of Adaptive Regrasp Planning

sparsity是什么无法理解。为什么训练EBM用200，sparity采样？训练的时候不用考虑sparsity吗？ 1.sparsity - 稀疏性；2. 好问题；最初我是这样想的，目前我们的方法可以仅考虑几何形状信息来生成大量的grasp candidiates。但有些candidiates因为物体质量分布不均匀，接触面积较小等，实际上无法完成稳定抓取（抓不住、在parallel gripper手内滑移），即稳定的抓取集合是当前采样抓取几何的一个子集。但实际情况下，我们没有理由不使用上全部采样到的grasp candidiates来训练E_{ϕ_f},来使得E_{ϕ_f}能够表征一个完整的可行抓取分布情况(仅考虑几何信息）。至于实际规划的时候用到那些抓取子集，其实都包含到了采样得到的集合当中，也能够进行合理的规划。那你就不应该用子集，而是从新采样50,100,150，200个 呃我感觉效果可能不会很好...，因为重新采样到的grasp会耦合对unseen grasp的泛化，实验结果在表格 II已说明。在这个实验中是假设训练用的200个包括了要用到的稳定抓取，只考虑子集也体现了稀疏性。感觉似乎也比较符合实际情况，stable grasp要从grasp candidates中一个一个让机器人尝试去实际抓取，然后筛选。用实际上不稳定的学习有意义吗?

This experiment evaluates the effectiveness of the proposed Iterative Deepening Planning framework (Section III-C) under varying levels of grasp sparsity. In practical applications, the number of effective grasps is often constrained by object geometry or kinematic limits, which can significantly weaken state space connectivity. We simulate different grasp sampling densities by subsampling a candidate grasp set $\mathcal{G}_{\text{sub}} \subset \mathcal{G}$, where $N_{\text{sub}} := |\mathcal{G}_{\text{sub}}|$ denotes the number of sampled grasps. For each object, we trained the EBM on 200 grasps and tested with $N_{\text{sub}} \in \{200, 150, 100, 50\}$. We conducted 10 independent experiments per sparsity level using different random seeds, each evaluating 10 pose pairs same as the previous experiment, yielding 100 episodes per object. Each episode sampled 1000 intermediate trajectories and optimized the K_{top}-10 and 50 candidates. We report Planning Success Rate (S_p, optimizer convergence) and Verification Success Rate of hard check (S_v, geometric feasibility via $\mathcal{S}_i \neq \emptyset$); gaps between S_p and S_v indicate convergence to infeasible local minima. Results are in Table IV and Fig. 8.

TABLE IV
STATISTICAL PERFORMANCE OF MULTI-STEP REGRASP PLANNING UNDER DIFFERENT GRASP SPARSITY LEVELS (N_{SUB}).K_{top}-10/K_{top}-50表格内容融合在一起了。

Object	N_{sub}	Success Rate (%) S_p	Success Rate (%) S_v	Efficiency $\bar{N}$	Efficiency $\bar{T}$ (s)
Bottle	200	89.0 / 93.0	89.0 / 93.0	1.20 / 1.06	0.82 / 0.67
	150	90.0 / 94.0	90.0 / 94.0	1.22 / 1.04	0.64 / 0.45
	100	82.0 / 89.0	82.0 / 89.0	1.34 / 1.03	0.54 / 0.32
	50	76.0 / 73.0	75.0 / 73.0	1.21 / 1.05	0.25 / 0.18
Mug	200	91.0 / 93.0	91.0 / 93.0	1.14 / 1.10	0.79 / 0.72
	150	72.0 / 63.0	67.0 / 62.0	1.19 / 1.10	0.69 / 0.55
	100	36.0 / 36.0	35.0 / 35.0	1.08 / 1.08	0.36 / 0.38
	50	22.0 / 19.0	22.0 / 19.0	1.09 / 1.05	0.21 / 0.20
Bunny	200	90.0 / 89.0	80.0 / 79.0	1.04 / 1.16	0.62 / 0.82
	150	90.0 / 90.0	80.0 / 79.0	1.12 / 1.11	0.58 / 0.52
	100	89.0 / 92.0	78.0 / 78.0	1.07 / 1.18	0.34 / 0.43
	50	81.0 / 77.0	76.0 / 64.0	1.14 / 1.16	0.22 / 0.25
Pentagon	200	100.0 / 99.0	89.0 / 88.0	1.04 / 1.04	0.60 / 0.58
	150	100.0 / 100.0	81.0 / 81.0	1.07 / 1.08	0.47 / 0.52
	100	97.0 / 98.0	72.0 / 72.0	1.07 / 1.08	0.33 / 0.38
	50	81.0 / 78.0	55.0 / 53.0	1.07 / 1.09	0.19 / 0.21

Note 1 Values are formatted as K_{top}-**10**/K_{top}-**50**.
Note 2 N_{sub}: Grasp sparsity level (subset size of candidates).
Note 3 S_p/S_v: Planning / Verification Success Rate over 100 episodes.
Note 4 $\bar{N}$: Average sequence length (step count) for successful cases.

The results indicate that when grasp candidates are sufficient ($N_{\text{sub}} = 200$), our planner effectively exploits an informative pose sequence cost landscape, leading to high planning success rates. However, as N_{sub} decreases, the connectivity weakens, revealing a clear performance divergence. For objects with complex geometry (e.g., Bunny), the planner maintains acceptable performance even at lower sparsity. In sharp contrast, for geometrically simple objects such as the Mug and the Pentagon, the reduction in grasp candidates critically undermines the solution space. As N_{sub} drops, the richness of the remaining candidates becomes insufficient to maintain global connectivity across the configuration manifold. Consequently, both planning (S_p) and verification (S_v) success rates suffer a precipitous decline. This confirms that for complex geometries, a dense grasp set is a prerequisite for ensuring the existence of valid regrasp sequences. Furthermore, our multi-step strategy consistently outperforms the one-step baseline across the same test pairs in Section IV-B. In scenarios where a single-step regrasp is strictly infeasible due to sparse connectivity, the framework restores solvability by autonomously introducing an intermediate pose. This transition from unsolvable to solvable underscores the effectiveness of the iterative deepening approach in navigating constrained configuration manifolds. Furthermore, comparing K_{top}-10 and K_{top}-50 reveals that a larger candidate pool does not necessarily degrade efficiency. In fact, K_{top}-50 often yields a lower average step count ($\bar{N}$), as a broader search space allows the optimizer to identify more direct paths, effectively bypassing redundant intermediate regrasps.

V. Conclusions and Future Work

This study elucidates the profound coupling between object geometry, end-effector performance, and grasp sampling distributions in regrasp planning. Experimental results demonstrate that increasing grasp sampling density and incorporating adaptive step-size planning significantly enhance manifold connectivity under complex constraints, thereby improving planning success rates.

Nevertheless, current planning efficiency still exhibits an inherent dependence on the density of the initial discrete grasp set. Even with an informative gradient field, the upper bound of planning robustness remains limited by the informational completeness of the sampling space, suggesting that representation learning alone cannot fully compensate for the connectivity gaps inherent in discrete sampling. Future research will focus on a paradigm shift from discrete candidate set searching toward continuous manifold generation.

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Appendix

A. Truncated Free Energy Score

Intuitively, when calculating the pose connectivity score Q_p between object poses $\mathbf{T}_a$ and $\mathbf{T}_b$, only those grasps that are simultaneously feasible in both configurations (i.e., shared grasps) contribute substantially to connectivity. To mitigate the noise introduced by high-energy infeasible grasps, we define the truncated pose connectivity score Q_p^h by restricting the free-energy summation to candidate grasps satisfying a joint energy threshold h. For brevity, denote $E_a(\boldsymbol{g}) = E_{\phi_f}(\mathbf{T}_a, \boldsymbol{g})$ and $E_b(\boldsymbol{g}) = E_{\phi_f}(\mathbf{T}_b, \boldsymbol{g})$:

$$
\begin{aligned}
Q_p^h(\mathbf{T}_a, \mathbf{T}_b) = -\alpha \log \sum_{\boldsymbol{g} \in \mathcal{G}} & \mathbb{I}[E_a(\boldsymbol{g}) + E_b(\boldsymbol{g}) < h] \\
& \cdot \exp\left(-\frac{E_a(\boldsymbol{g}) + E_b(\boldsymbol{g})}{\alpha}\right).
\end{aligned}
\tag{10}
$$

The truncation threshold h is determined statistically. We evaluated the model on a pre-trained shared grasp validation set and selected the energy boundary that maximizes the F_1 score for the shared grasp classification task as h. When $\mathcal{G}_{\mathrm{shared}} = \emptyset$, the absence of feasible samples creates a gradient-free 'null' state, assigned a constant penalty of 0. This plateau is intrinsic to truncation: excluding sub-optimal grasps removes the directional cues required to guide optimization through infeasible regions, leading to the vanishing gradients shown in Fig. 5(b.ii).

B. EBM Training Details

We train the energy function $E_{\phi_f}(\mathbf{T}, \boldsymbol{g})$ to assign lower energy to feasible grasps and higher energy to infeasible ones. The training objective combines three loss terms: (1) Negative Log-Likelihood (NLL) loss that encourages low energy for feasible samples while normalizing via the partition function; (2) Contrastive loss that explicitly separates feasible and infeasible grasps; and (3) Regularization loss that prevents energy divergence. The final loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{nll}} + \mathcal{L}_{\text{con}} + \alpha\mathcal{L}_{\text{reg}}$, where α is a weighting factor.

Training data is collected following [25]: grasp candidates are generated by sampling antipodal contact pairs on object surfaces, and object poses are sampled from stable placements diversified by planar translations and rotations. The EBM is implemented as a three-layer fully connected neural network with SELU activation functions. During inference, grasps are classified as feasible if their energy falls below a threshold h_f determined via validation set F1-score.